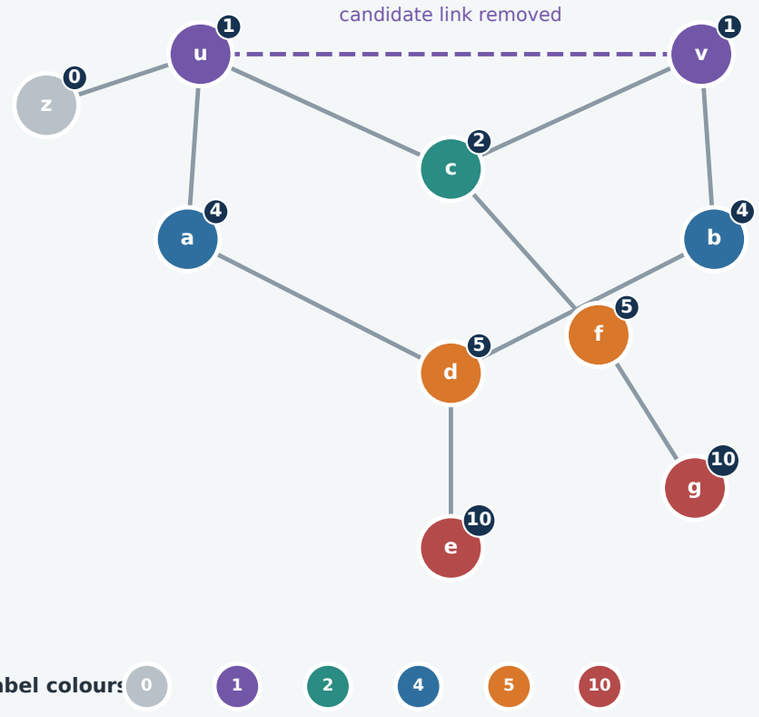
DRNL | Detailed worked example

Distances are measured independently to each target; the pair is compressed into one discrete structural label

A. Enclosing-subgraph example

Badge = computed DRNL label z(i). The dashed target edge is excluded from the classifier input.



B. Exact mapping

$d_u(i)$: shortest path from i to u after removing v

$d_v(i)$: shortest path from i to v after removing u

Special cases
z(u) = z(v) = 1
∞ distance → z(i) = 0

$$d = d_u + d_v, \quad q = \lfloor d/2 \rfloor, \quad r = d \bmod 2$$

$$z(i) = 1 + \min(d_u, d_v) + q\,(q + r - 1)$$

C. Node-by-node calculation

Node i	$d_u(i)$	$d_v(i)$	q	r	z(i)
a	1	3	2	0	**4**
b	3	1	2	0	**4**
c	1	1	1	0	**2**
d	2	2	2	0	**5**
e	3	3	3	0	**10**
f	2	2	2	0	**5**
g	3	3	3	0	**10**
z	1	∞	—	—	**0**

MODEL INPUT integer label z(i) → one-hot DRNL node tag → DGCNN graph convolutions

Topology becomes a learnable categorical feature

Formula and edge cases validated against src/python/util_functions.py